

Malitha Gunawardhana

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SUMMARY

Machine Learning, Data Science, and Software Engineering professional with experience building and deploying applied AI, predictive systems and data solutions, plus a PhD in deep learning completed in August 2026. Experienced across the end-to-end ML lifecycle, from data pipelines and curation through model development, evaluation and production deployment, using Python, PyTorch, SQL and cloud/MLOps tooling. Track record of measurable production impact, including a 70% reduction in manual processing time, with experience translating ambiguous, cross-disciplinary requirements into systems used by technical and non-technical stakeholders.

EXPERIENCE

Machine Learning Engineer (Part-Time)

March 2026 - Sep. 2026

Viotel Pty Ltd

Parnell, Auckland

- Developed and deployed a deep-learning system for **seismic wave phase identification** across **3,700+ seismic events and 30,000+ station samples**, cutting manual processing time by **more than 70%**.
- Developed and deployed **Python time-series and anomaly-detection pipelines** that flag abnormal vibration patterns across **8+ telecommunications towers**, replacing manual inspection with automated monitoring of live sensor feeds.
- Conducted **data-quality and diagnostic analysis** on raw sensor streams with geophysicists and a three-member field team, isolating instrument faults, validating model outputs against expert labels, and reporting findings that guided deployment of **17 seismic sensors** across the Taupō region.

Machine Learning Engineer (remote from Dec. 2023)

Sep. 2022 - Jan. 2025

Institute of Fundamental Technological Research, Polish Academy of Sciences (IPPT PAN)

Warsaw, Poland

- Developed deep-learning models for **breast tumour identification from ultrasound images**, reaching **86% identification accuracy** and enabling a low-cost computer-aided detection system for resource-limited screening settings.
- Designed **reusable, reproducible medical-imaging pipelines** covering data sourcing, curation, preprocessing, quality assurance, training, evaluation and cross-dataset comparison, enabling new datasets to be incorporated without rebuilding the workflow.
- Collaborated with **clinicians, researchers and engineers** to define requirements, establish evaluation criteria, communicate model limitations, and keep technical outputs usable by non-ML stakeholders.

Artificial Intelligence Researcher (concurrent with IPPT PAN role)

Sep. 2022 - Sep. 2023

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)

Abu Dhabi, UAE

- Developed **self-supervised and semi-supervised** models for videos and images that maintained performance under limited labelled-data conditions, reducing the volume of manual annotation required.
- Conducted controlled experiments on **LLMs, network calibration and deep-learning reliability**, quantifying failure modes and producing evaluation evidence that informed model-selection decisions.
- Led a four-member team through experiment design, analysis and delivery of findings, with results published and presented at leading artificial intelligence conferences.

Machine Learning Engineer

June 2022 - Nov. 2022

PromiseQ GmbH

Berlin, Germany

- Optimised deep-learning models for a **live surveillance platform running across 3,000+ CCTV cameras**, diagnosing recurring failure modes and prediction errors in production traffic.
- Reduced false alarms by 5%** through network calibration and error analysis, measurably improving the reliability of a tool used by operators every day.
- Collaborated across engineering and ML teams to monitor deployed-model performance, triage production incidents, and improve system reliability.

Full-Stack Software Engineer

March 2021 - May 2022

Xeptagon (Pvt) Ltd

Colombo, Sri Lanka

- Engaged directly with senior stakeholders, including **CEOs, CTOs and managers**, to understand operational processes, clarify ambiguous and evolving requirements, and translate them into production systems.
- Led a four-member team delivering a low-latency **domain drop-catching system** with over **90% success** at under **50 ms latency**, integrating concurrent backend components and optimising execution performance.
- Developed audio-processing functionality for a student learning management system, transforming **80+ audio signals** into reusable structured features through signal processing and feature engineering.

Software Engineer (Intern)

June 2019 - Dec. 2019

Synergen Technology Labs (Pvt) Ltd

Colombo, Sri Lanka

- Contributed to the design and development of a wearable physiological-signal acquisition device, and collected an original dataset through controlled stress-inducing experiments.
- Developed physiological signal-processing and ML pipelines for stress classification, achieving over **90% accuracy** across four stress categories.

EDUCATION

Doctor of Philosophy (Deep learning, Computer Vision & Data Science)

Dec. 2023 - Aug. 2026

University of Auckland

Auckland, New Zealand

- Thesis: Robust, Uncertainty-Aware, and Scalable Deep Learning for Bi-Atrial Segmentation from LGE-MRI in Atrial Fibrillation. **Award: Health Research Council Scholarship.** Findings published and presented at international conferences.
- Designed and conducted controlled experiments on **semantic segmentation, self-supervised learning and network calibration** across multiple large imaging datasets, applying **cross-validation, statistical hypothesis testing, uncertainty quantification and error analysis** to compare competing approaches and communicate results to clinical and engineering stakeholders.
- Managed the end-to-end data workflow, including **data curation, cleaning, preprocessing, exploratory data analysis and label quality assurance**, resolving distribution shift and annotation inconsistency across sources.
- Developed reproducible **Python ML pipelines with experiment tracking and version control**, enabling systematic **benchmarking, cross-dataset validation and failure analysis**.
- Developed a **RAG-based literature agent** over a local corpus of research papers using PDF parsing, chunking, **embeddings and vector search** with an LLM agent, answering natural-language queries with cited source passages and streamlining literature retrieval.

B.Sc. Engineering Honours

Jan. 2017 - July 2021

University of Moratuwa

Moratuwa, Sri Lanka

- Dean's list placement in Semester 7. Key modules: Engineering Mathematics, Statistics, Image Processing and Machine Vision, Neural Networks and Fuzzy Logic, Medical Imaging, and Signal Processing.

Certifications & Training

- **Oracle Cloud Infrastructure 2024 Generative AI Certified Professional** - LLM fundamentals, pre-training, fine-tuning and alignment, generative AI, MLOps.
- **Microsoft Certified: Azure Fundamentals (AZ-900)** - Azure cloud services, storage, networking, security and virtualisation.
- **Large Language Model Agents**, UC Berkeley - LLM agents, multi-agents, RAG, MLOps.
- **Data Science Career Track**, 365 Data Science - Python, SQL, data wrangling, statistical analysis, model evaluation.

TECHNICAL SKILLS

Programming Languages: Python (Pandas, NumPy), SQL, Bash, TypeScript, MATLAB, R

Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, Transformers, SSMs, Computer Vision, Deep Learning, Predictive and Time-Series Modelling

Mathematics and Statistics: Linear Algebra, Probability, Optimisation, Hypothesis Testing, Cross-Validation, Uncertainty Quantification

Generative AI and LLMs: RAG, LangChain, CrewAI, Weaviate, LLM and Agent-Based Systems, Agentic Workflows, Prompt Engineering, Claude, Codex

Data and MLOps: MLflow, Apache Airflow, PySpark, Kafka, Weights & Biases, Neptune, Docker, Jenkins CI/CD

Cloud and Engineering: AWS, Azure, GCP, Git, Linux, REST APIs, MongoDB, Jira, Confluence

Data Analytics and BI: Power BI, Tableau, Microsoft Excel, Statistical Analysis, Data Visualisation

SERVICE AND LEADERSHIP

Auckland Bioengineering Institute, University of Auckland

Feb. 2024 - Aug. 2026

- Elected Student Council Member representing postgraduate students in university governance, student advocacy, welfare initiatives and policy discussions.

Sustainable Education Foundation

July 2024 - Apr. 2025

- Mentored students through structured academic, professional and personal development guidance.

Rotaract Club of Alumni of the University of Moratuwa

July 2022 - July 2024

- Vice President for Club Service, coordinating member engagement, fellowship activities and community initiatives.